

Propellantless electron thruster for LEO station keeping

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Abstract

Small spacecraft below 600 km lose their orbit to atmospheric drag within months to a few years, and most of them carry no propulsion at all. This paper describes a thruster with no tank, no feed system and no neutralizer. A cathode held a few hundred volts below the spacecraft body emits electrons through a single aperture. The escaping beam produces thrust and leaves the body positively charged, so the body collects electrons from the ionospheric plasma until collection equals emission. The circuit closes through the ionosphere with zero net mass flow. Particle-in-cell simulations (WarpX) of a 10 mm conducting body in a dayside ionospheric plasma measure the complete operating envelope over 100–350 V: 3.4 to 43 nN of thrust from 12 to 278 mW of beam power, with 96–99 % of the beam escaping, a floating potential of 4–48 V, and an energy efficiency of 0.68–0.77. Two measured constants describe every run to within 3 %. Against a year of real 2024 drag from orbit propagation, with the supply voltage following the drag and the floating potential solved from the plasma along the orbit, the device holds 500–600 km for a mean beam power of 8–39 mW and stays inside the tested 100–350 V envelope more than 99 % of the time; at 400 km it needs 140–240 mW, and the drag peaks call for 500–700 V, beyond the simulated envelope. Because both drag and electron collection scale with area, the feasibility condition is independent of vehicle size, and a 3U CubeSat needs 0.9–3.4 W for the same altitudes. A bench experiment confirms the electrode topology on hardware. A three-dimensional follow-up campaign puts the geomagnetic field perpendicular to the beam, the orientation the mission flies in: at flight strength the operating point survives (thrust within 0.8 %, escape unchanged, a 2.6 V float shift at the settled equilibrium), while at ten times flight strength the return circuit cannot close and the device must fire along the field. Open items are the choice of flight emitter and the CubeSat sheath regime.

1 Introduction

Small spacecraft below 600 km lose their orbits to atmospheric drag within months to a few years. A 3U CubeSat at 400 km falls below 100 km in roughly 139 days. Most carry no propulsion at all.

The thrust to hold altitude is nanonewtons to a few micronewtons, well within the range of flight-proven micropropulsion. Colloid thrusters on LISA Pathfinder resolved 0.1 μ N [1]; NanoFEEP flew on a 1U CubeSat at 0.2–0.3 W [2]. Every one of these systems spends a stored propellant through tanks, valves, feed lines or precision emitter heads. The propellant imposes a total-impulse limit, and the hardware adds mass and cost that are rarely justified on a CubeSat.

This paper describes a thruster with none of that hardware. A cathode held below the spacecraft potential emits electrons through an aperture. The escaping beam produces thrust and leaves the body positively charged, so the body collects ambient electrons from the ionospheric plasma until collection balances emission. The circuit closes through the ionosphere with no propellant and no

net mass flow. An electron beam carries about 200 times less momentum per watt than an ion beam, but LEO drag on a small spacecraft is nanonewtons, and at that scale the beam costs tens of milliwatts. The whole device is a cathode, an aperture and a high-voltage supply. It carries no propellant, no tank and no feed system, and at 400–600 km it produces enough thrust for station keeping or extending mission life.

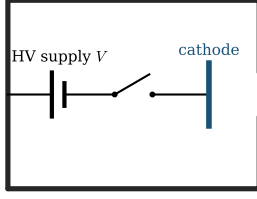
The following sections describe the operating principle, test each claim through a ladder of particle-in-cell simulations that build from a vacuum electron gun to the full device in ionospheric plasma, and assess the result against a year of real 2024 drag at 400–600 km.

2 Concept

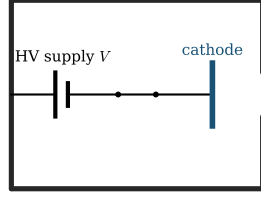
The device has three parts (Fig. 1):

1. a cathode, held at $-V$ with respect to the spacecraft body, that emits electrons and accelerates them out through an aperture;
2. the spacecraft's own conducting skin, which is the

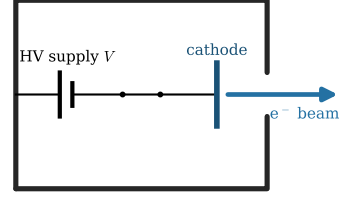
1. initial state



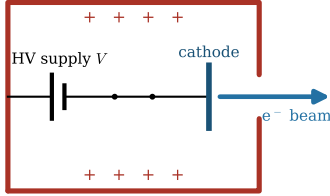
2. supply switched on



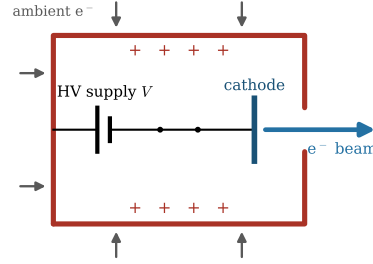
3. cathode emits, beam accelerated out



4. body charges positive



5. positive body collects ambient e-



6. steady state: collection = emission

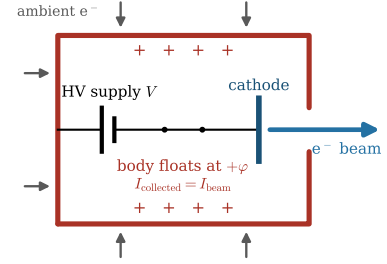


Figure 1: Operating sequence. The supply is switched on, the cathode emits, the escaping beam charges the body positive, and the positive body collects ambient electrons until collection equals emission.

second electrode and collects electrons from the plasma;

3. a high-voltage supply between the two, which moves collected electrons back to the cathode.

Once switched on, the device reaches a steady state:

1. the spacecraft floats in the ionospheric plasma;
2. the supply is switched on and the cathode emits;
3. each escaping electron leaves the body one charge more positive;
4. the positive body attracts electrons from the plasma;
5. collection grows until it equals emission. The body settles at a potential φ where the two currents balance.

Two questions decide whether this works.

Why does the positive body not recapture its own beam? Because the beam electrons carry 100–300 eV while φ is a few volts to a few tens of volts; ambient electrons at 0.1 eV are captured, beam electrons are not.

Does the returning current cancel the thrust? The collected electrons gain 4–48 eV from the floating potential, against 77–272 eV for the beam, and they arrive from all directions rather than straight astern. The simulations measure the net cancellation at 0.4–1.4 % of the beam thrust at 200–350 V, rising to 9 % at 100 V where the slower beam carries less momentum relative to the collected electrons (Sec. 4.4.1).

3 Theory

3.1 Thrust

The beam is a current I of electrons with kinetic energy KE. Its momentum flux is the thrust:

$$F = \frac{I\sqrt{2m_e \text{KE}}}{e} = 3.37 \frac{\text{nN}}{\text{mA}\sqrt{\text{eV}}} I\sqrt{\text{KE}}. \quad (1)$$

The constant contains only the electron mass and charge. The exhaust velocity is $v_e = \sqrt{2e \text{KE}/m_e}$, of order 10^7 m/s.

3.2 The float tax

The body floats at $+\varphi$ and the cathode at $\varphi - V$, so the accelerating potential is $V - \varphi$. Space charge near the cathode depresses it further. The exhaust energy is

$$\text{KE} = \kappa (V - \varphi), \quad (2)$$

where κ is the gun's energy transmission, set by how hard the gun is loaded relative to its space-charge limit. The ratio φ/V is the fraction of supply voltage spent on collecting the return current.

3.3 Power and efficiency

Beam power is $P = VI$. Jet power over supply power is

$$\eta = \kappa \frac{V - \varphi}{V} f_{\text{esc}}, \quad (3)$$

with f_{esc} the fraction of the beam that clears the body. Eliminating I between Eq. (1) and $P = VI$ at fixed thrust gives $P \propto F\sqrt{V}$: doubling the voltage costs about

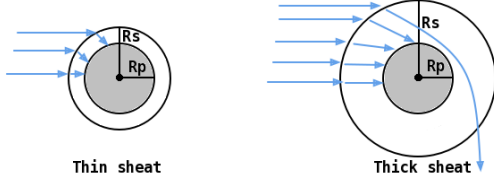


Figure 2: Thin sheath ($r_p \gg \lambda_D$): electrons reach the body on nearly straight paths and collection scales with surface area. Thick sheath ($r_p \lesssim \lambda_D$): the potential extends far beyond the body and bends electron orbits inward, giving the $(1 + \chi)^\alpha$ enhancement of Eq. (5). The 10 mm body sits in the thick-sheath regime; a CubeSat would sit in the thin-sheath regime.

40 % more power for the same thrust. Thrust per watt is $F/P \approx 3.37\sqrt{\kappa}/\sqrt{V}$ in $\mu\text{N/W}$ with V in volts, about $0.2 \mu\text{N/W}$ at 200 V.

3.4 Emission ceiling

Low voltage is cheap, but the current a gun can source is limited by space charge. The planar Child–Langmuir scale for the simulated spot ($r = 0.5$ mm, gap 4.7 mm) is $I_{\text{CL}} \propto V^{3/2}/d^2$, equal to 0.08, 0.24 and 0.43 mA at 100, 200 and 300 V; the real gun emits 1.46 times that. Maximum thrust therefore scales as $I_{\text{CL}}\sqrt{V} \propto V^2$, and at 100 V the spot sources about 0.12 mA whatever the plasma does. In the $\varphi \ll V$ limit the best operating point is the lowest voltage whose emission ceiling still meets the thrust demand; with the float included, the minimum-power voltage sits somewhat higher (Sec. 5). A shorter gap or a wider spot moves the ceiling without changing the voltage.

3.5 Current collection

The body collects the one-sided thermal electron flux [6]

$$I_{\text{th}} = e A n_e \sqrt{\frac{kT_e}{2\pi m_e}}, \quad (4)$$

enhanced by the positive potential. With $\chi = e\varphi/kT_e$,

$$I = \beta I_{\text{th}} (1 + \chi)^\alpha, \quad (5)$$

where $\alpha = 1$ for a sphere and 0.5 for a long cylinder in orbital-motion-limited theory, and β is a geometry factor. Setting I equal to the escaping current and inverting gives the float. Its stiffness goes as $1/(n_e\sqrt{T_e})$, so thin night-side plasma makes the body float higher for the same current.

3.6 Scale

Drag is proportional to the ram area; collection and body-mounted power are proportional to the skin area. Thrust demand, beam current and power all scale with ram area, so power per unit ram area depends only on altitude and voltage. Vehicle size drops out and what remains is the

shape ratio $A_{\text{skin}}/A_{\text{ram}}$. A 10 mm can and a 3U CubeSat with the same shape ratio have the same feasibility margin.

3.7 Control

I and φ are plain electrical measurements, so Eqs. (1)–(2) let the spacecraft compute its own thrust in flight. Density, temperature and day/night all collapse into where the body floats, so a two-line servo on the measured float is the whole control law. A microcontroller is sufficient.

4 PIC simulations

4.1 Setup

The ladder and characterization runs use WarpX [3] (built on AMReX [4]) in RZ electrostatic mode with the parameters of Table 1; the transverse-field campaign of Sec. 4.4 uses the same code in Cartesian 3D. The body is a closed can with a cathode disk inside and an aperture in the lid, modelled as an embedded conductor (Fig. 3). Its potential is not prescribed. Each step the code counts the charge that left through the aperture and the charge that landed on the skin, and moves the body potential by the net charge over the body’s self-capacitance, which is measured once from Gauss’s law on a vacuum Laplace solve. The float is therefore an output, and ladder stage 7 checks the method on a bare body against the theoretical floating potential. Ions are repelled by the positive body, so the balance is between the escaping beam and collected electrons only; the reduced ion mass affects how fast the sheath settles, not where it settles. The ambient plasma is a dayside 400 km row kept constant by a particle reservoir. Every run freezes its configuration and passes fixed acceptance gates on escape fraction, floating potential, current balance and momentum before it is used.

4.2 Validation ladder

A PIC simulation of the full device has too many parameters to validate at once. The ladder isolates each piece of physics and checks it against theory before combining them, so the full thruster result inherits confidence from every stage below it. Nine stages build from a vacuum electron gun to the full device (Table 2). All nine pass.

Stages 1–3 test the gun alone in vacuum. Stage 1 confirms that a negative cathode emits and accelerates electrons through a near-neutral aperture, the basic electrode topology, with a $35 \mu\text{V}$ energy error on 100 V; the same topology is confirmed on hardware in Sec. 7. Stage 2 measures aperture transmission at three hole sizes ($\kappa = 0.97, 0.90$ and 1.00 , matching the predicted values). Stage 3 confirms transmission is independent of voltage

Table 1: PIC configuration of the full thruster at 200 V. Characterization runs change one entry at a time (Table 4).

	parameter	value
geometry	body outer diameter $\times$ length	10×5.5 mm
	wall thickness	0.4 mm
	exhaust aperture diameter	4 mm
	cathode disk diameter	3 mm
	emission spot diameter	1 mm
	cathode–lid gap	4.7 mm
electrical	supply voltage V	200 V
	beam current I (prescribed)	0.342 mA
	gun loading I/I_{CL}	1.46
	body self-capacitance	0.5–1 pF
plasma	n_e	$1.63 \times 10^{12} \text{ m}^{-3}$
	kT_e	114 meV
	kT_i	81 meV
	ion mass	$400 m_e$
numerics	cell size $\Delta r = \Delta z$	0.15 mm
	domain $r_{\max}$	30 mm
	particles per cell per species	16
	run length	800 ns (29 RC times)
	supply / gun switch-on	100 / 150 ns

Table 2: Validation ladder.

#	stage	validates
1	negative cathode	emits through neutral aperture
2	holed anode	aperture controls transmission
3	voltage bracket	transmission is V -independent
4	thermal collection	PIC current vs Eq. (4)
5	biased, +3 V	OML collection at bias
6	biased, +10 V	larger sheath, more current
7	floating body	unbiased float vs theory
8	two-node Laplace	two potentials, one body
9	full thruster, 200 V	gun + plasma combined

(0.006 pp spread over 200–300 V). The transmissions in stages 2–3 are for lightly loaded vacuum guns; the device gun runs at 1.46 times its space-charge scale, which is where the lower $\kappa = 0.81$ of Sec. 4.4.1 comes from.

Stages 4–7 test collection on a conducting body in the plasma. Stage 4 reproduces the thermal electron flux of Eq. (4) to within 1 %. Stages 5–6 bias the body to +3 and +10 V and measure 0.85 and 0.81 of the Laframboise OML ceiling, with the sheath growing from 4.1 to 6.9 mm as the potential rises. Stage 7 lets the body float freely and measures -0.251 V, inside the bracket between two theoretical models. Stage 8 places two prescribed potentials on one embedded conductor and recovers the exact Laplace solution with zero violation, which validates the multi-potential method used in stage 9. Stage 9 is the full thruster, described next.

Table 3: Steady-state values of the full thruster at 200 V. This run is the anchor for all characterization.

quantity	value
floating potential φ	17.0 V
beam escape f_{esc}	98.4 %
thrust F	13.65 nN
beam power P	68.4 mW
exhaust energy KE	147.5 eV
energy efficiency η	0.73

Table 4: Characterization runs. Each changes one axis of the full thruster except the last, which combines two.

axis	φ [V]	F [nN]	f_{esc} [%]	KE [eV]
full thruster (200 V)	17.0	13.65	98.4	147.5
100 V	5.4	3.42	96.1	77.2
300 V	36.3	30.13	99.0	210.1
350 V	48.3	40.48	99.1	239.0
slender body ($L/r = 6$)	4.4	14.22	98.4	159.7
density $n_e/3$	42.5	12.39	99.1	122.0
$B_z = 30 \mu\text{T}$	17.2	13.64	98.4	147.3
$B_z = 300 \mu\text{T}$	48.6	12.06	98.3	115.9
350 V + slender	14.0	43.33	99.1	272.7

4.3 Full thruster

Stage 9 puts the gun inside the floating can and switches it on in the 400 km plasma (Fig. 3). The body charges positive as beam leaves, collection rises with the float, and the two currents balance within a few RC times. Table 3 lists the steady state at 200 V. The body floats at 17 V, 98.4 % of the beam escapes, and the thrust is 13.65 nN at 68.4 mW of beam power. This run is the anchor: everything in the next section perturbs it one axis at a time.

4.4 Characterization

Eight runs vary the full thruster one axis at a time: voltage first, then shape, plasma density and magnetic field, and one run that combines two axes (Table 4). All pass their gates. Predictions were written down before each run.

4.4.1 Voltage envelope

Three runs at 100, 300 and 350 V hold the same fraction of the Child–Langmuir current and the same plasma (Table 5, Fig. 4). Pre-registered thrust predictions landed within 1 %.

Two constants fit all four points:

$$F [\text{nN}] = 3.27 I [\text{mA}] \sqrt{\text{KE} [\text{eV}]}, \quad (6)$$

$$\text{KE} = 0.81 (V - \varphi). \quad (7)$$

The thrust slope is 97 % of Eq. (1); the deficit is plume divergence. $\kappa = 0.81$ is the space-charge depression of a single-aperture gun at this loading. Thrust per watt follows $1/\sqrt{V}$ within 2.5 % over 100–300 V, and the 350 V point sits 3 % below it because the float tax has grown to 14 %. The energy efficiency falls slowly from 0.74 to

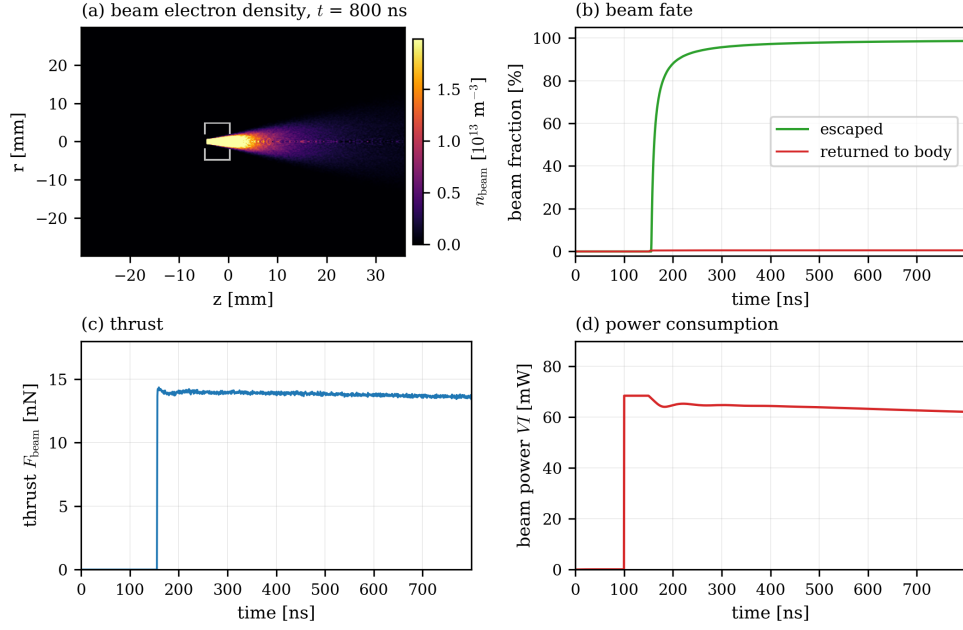


Figure 3: The full thruster at 200 V, the reference run for everything that follows. (a) Beam electron density at the end of the run; the grey outline is the can. (b) Fraction of emitted beam that has escaped the domain or returned to the body. (c) Beam thrust. (d) Drive power left after the float, $(V - \varphi) I$. The gun switches on at 150 ns; thrust, power and escape are all steady well before 800 ns.

Table 5: Measured operating envelope of the 10 mm body. Grid-resolution uncertainty: $F \pm 4\%$, $\text{KE} \pm 7\%$, $\varphi \pm 2\%$.

V V	I mA	φ V	F nN	P mW	f_{esc} %	η
100	0.121	5.4	3.42	12.1	96.1	0.74
200	0.342	17.0	13.65	68.4	98.4	0.73
300	0.630	36.3	30.13	189	99.0	0.69
350	0.793	48.3	40.48	278	99.1	0.68

0.68 as the float tax grows; almost every electron leaves at nearly full energy.

The 300 V run also selects the collection exponent. Three values of α in Eq. (5) agree at 200 V and predict 31, 36 and 90 V for the 300 V float; the measurement is 36.3 V. A fit across the envelope gives $\alpha = 0.82$ – 0.89 , between the sphere and cylinder limits. The 350 V float was predicted from that fit at 47 V and measured at 48.3 V. One caveat applies to both high-voltage points: the sheath settles on the ion timescale, and their floats were still rising slowly at run end. The quoted values are tail averages; extrapolating the decaying slope puts the settled 300 V float at 42–48 V and the settled 350 V float past 50 V. No float value is a design limit; the float is a tax on the drive through Eq. (2), 14 % of the supply for the compact body at 350 V against 4 % for the slender body, which is why the slender body matters at 400 km.

4.4.2 Shape, density and field

Shape. Does elongation hurt current collection? Stretching the can to $L/r = 6$ triples the skin at the same ram area. Two hypotheses were recorded before the run: area-only scaling predicted 4–5 V, and the long-cylinder

limit predicted tens of volts. The float dropped to 4.4 V and the thrust rose 4 %, confirming area-only scaling. The collection exponent held across a $3.2\times$ area change. A 3U CubeSat flying end-on has the same shape ratio, so the measured exponent applies directly (Sec. 5.4).

Density. Does thin night-side plasma break the device? At one third density the run was extended to $2.4 \mu\text{s}$ (three times the RC time) and the float settled at 42.5 V. Every fixed-exponent prediction from Eq. (5) overshoot, from 53 V at $\alpha = 1$ to 68 V at $\alpha = 0.82$. The fitted law overestimates the float in thin plasma, so night-side predictions from Sec. 5 are conservative.

Magnetic field. Does the geomagnetic field tax collection? Along the thrust axis, no. An axial field at flight strength ($30 \mu\text{T}$) leaves float, thrust and escape within 0.2 V, 0.1 % and 0.06 percentage points of the reference run; at ten times flight strength collection is taxed, the float rises 33 V, and the thrust law absorbs the change once the measured float is inserted. The transverse orientation, the one the mission flies in, was then measured on a Cartesian 3D deck: the same body resolved at 1 mm cells in a ± 60 mm box, the escaped beam prescribed at the lid with the anchor’s measured energy, and the thrust ledger corrected for the Lorentz force on the particles in flight, since the exit flux alone under-reads the emission reaction once the beam curls. Runs of $6 \mu\text{s}$ settle the float on the model’s ion clock; the 800 ns snapshots reach about half the settled value, and thrust moves under 4 %. At flight strength the operating point survives: thrust within 0.8 %, escape unchanged, settled float 29.4 V against the control’s 26.8 V, a 2.6 V tax, about

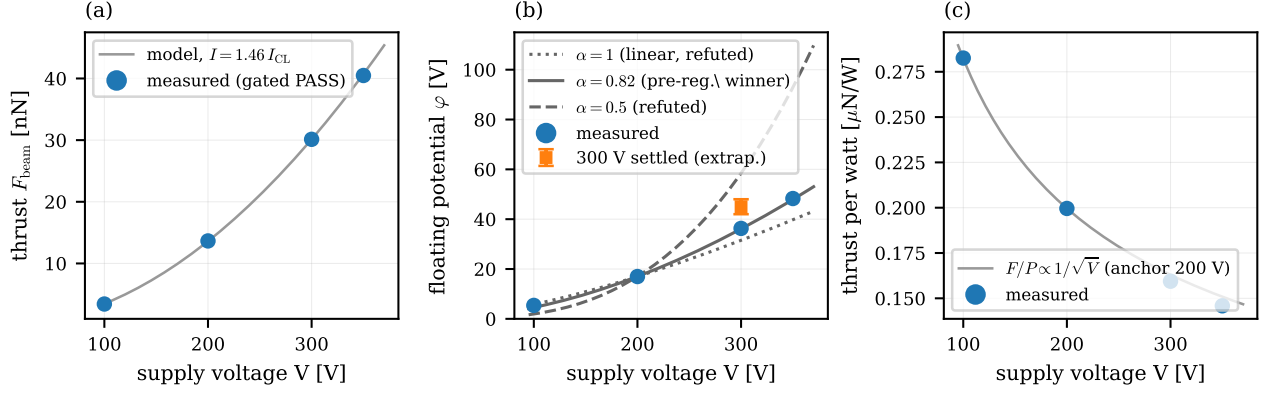


Figure 4: The four envelope runs (dots) against the model (lines). Left: measured thrust and the two-constant law $F = 3.27 I \sqrt{KE}$, $KE = 0.81(V - \phi)$. Centre: measured floating potential and the float predicted by Eq. (5) for three collection exponents, each forced through the 200 V point; they diverge above 200 V, and the 300 V measurement selects $\alpha \approx 0.82$ – 0.89 and refutes the sphere ($\alpha = 1$) and cylinder ($\alpha = 0.5$) limits. The square is the 300 V float extrapolated to sheath settlement. Right: measured thrust per watt and the $1/\sqrt{V}$ law that follows from $F \propto I\sqrt{V}$, $P = IV$, anchored at 200 V; the 350 V point falls below it because the float tax has grown.

1 % of the drive. At ten times flight strength the circuit does not close: the float climbs through the 150 V abort ceiling with no equilibrium below 91 % of the beam energy, so in amplified fields the device must fire along the field, the mode the axial runs validate.

400 km operating point. Does the slender body at 350 V close the 400 km mean drag (32.9 nN)? The voltage and shape laws together predicted 11–17 V and 42–43 nN. It measured 14.0 V and 43.33 nN, covering the mean demand with a float well inside the 50 V limit. The two laws compose, which turns two one-dimensional curves into an operating map over 100–350 V and $L/r = 0.6$ – 6 .

5 Mission assessment

5.1 Method

Drag demand comes from TudatPy [5] propagation over 2024 (solar-cycle maximum) with NRLMSISE-00 density driven by measured F10.7 and A_p , and IRI-2020 plasma along the orbit. The test body is the simulated 10 mm $\times$ 5 mm cylinder at 100 g with $C_d = 2.2$, on a circular near-equatorial orbit ($i = 0.5^\circ$) at 400 km (axial and lateral pose), 500, 550 and 600 km. Drag is cancelled exactly, so the drag at each timestep is the thrust demand. Table 6 summarizes the demand and the plasma each case sees. The PIC plasma row of Table 1 sits near the dayside top of the 400 km range; night-side densities are 5–40 times lower, which is where collection is weakest.

5.2 Coverage

At 500–600 km the thruster covers mean and maximum drag at a fixed 300 V (Fig. 5, Table 7). In voltage-

Table 6: Mission cases: 2024 drag on the 10 mm body and the IRI-2020 plasma along the orbit (5th–95th percentile of 5-min samples over the year).

altitude	pose	drag [nN]		$n_e [10^{11} \text{ m}^{-3}]$	$kT_e [\text{meV}]$
		mean	max		
400 km	axial	32.9	92.4	2.7–21	87–193
400 km	lateral	21.6	60.7	2.7–21	87–193
500 km	axial	7.6	28.4	1.4–16	91–221
550 km	axial	3.8	16.3	1.0–11	93–233
600 km	axial	2.0	9.6	0.7–7.1	97–244

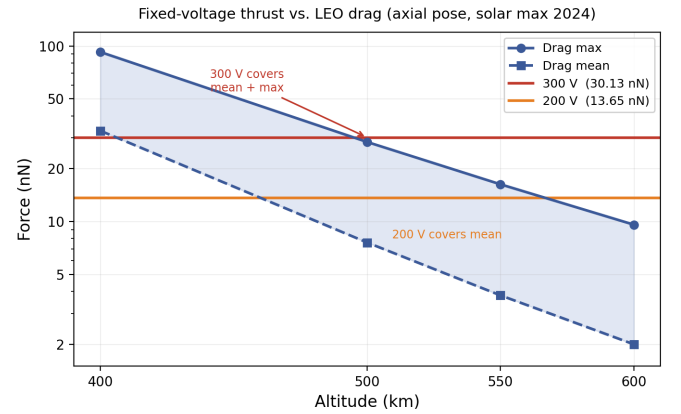


Figure 5: Measured thrust at 200 and 300 V against mean and maximum 2024 drag on the 10 mm body, by altitude.

following mode it holds the orbit all year for a mean beam power of 8–39 mW, and more than 99 % of the orbit points sit inside the tested 100–350 V envelope (Table 8). The float raises the year mean 20–25 % above the ϕ -free law of Sec. 3, because neutral density and plasma density fall together on the night side: the demand drops where collection is weakest, and the most expensive 5 min of the year at 600 km is a dayside drag peak at 55 mW. No voltage or float value is a design limit in this model; a high float on a thin plasma costs supply voltage and power at that point through Eq. (2), and every point of the 2024 year has an operating point. At 400 km

Table 7: Fixed-voltage coverage of 2024 drag on the 10 mm body.

altitude	mean nN	max nN	200 V 13.65 nN	300 V 30.13 nN
400 km axial	32.9	92.4	–	–
400 km lateral	21.6	60.7	–	mean
500 km	7.6	28.4	mean	mean, max
550 km	3.8	16.3	mean	mean, max
600 km	2.0	9.6	mean, max	mean, max

Table 8: Voltage-following mode: for each 5-minute point of the 2024 orbit, the supply voltage that cancels that point’s drag for the least beam power, with the floating potential solved from the IRI-2020 plasma at that point. Points above 350 V use the gun laws beyond the simulated envelope.

altitude	mean P	worst 5 min	$V > 350$ V	φ median/99th
600 km	8.3 mW	55 mW	0.02 %	18/61 V
550 km	17 mW	100 mW	0.1 %	23/81 V
500 km	39 mW	190 mW	0.8 %	28/110 V
400 km lateral	139 mW	611 mW	11 %	39/162 V
400 km axial	239 mW	1.08 W	40 %	56/219 V

the compact body delivers 40.48 nN at 350 V against a 32.9 nN mean and pays 14 % of the drive to the float; the slender body delivers 43.33 nN at a 4 % tax, covering the mean at about 76 % duty. The drag maxima (92 nN) need 500–700 V by the gun laws, above anything simulated, and the year costs 140–240 mW, so 400 km is a research target, not a closed case. Where full station-keeping does not close, partial compensation still extends mission life. Two parts of Table 8 are extrapolations, flagged per point in the model output: the collection law below about a third of the simulated density, where the one off-density run found it conservative, and the calibration on 800 ns floats (Sec. 8).

5.3 Power

Power is the thruster’s only interface to the spacecraft: tens of milliwatts at 500–600 km, 100–200 mW at 400 km. The 10 mm body harvests roughly 10–30 mW from body-mounted cells. A 1U CubeSat has 1–2 W and a 3U 5–10 W. Supply and demand both grow with area, so the ratio holds with size.

5.4 Scaling to CubeSats

By the argument of Sec. 3, a 3U CubeSat flying end-on has the same shape ratio as the slender can and the same feasibility margin. In absolute terms it needs 8.8 mA and 0.9 W at 600 km, 1.7 W at 550 km and 3.4 W at 500 km. Larger bodies also need less collection enhancement over the thermal flux (4 times for a 3U at 600 km against 15 times for the 10 mm body), so they rest on less extrapolation, not more. These numbers are estimates from the fitted collection law, and CubeSat radii sit at 25–60 Debye lengths where that law has not been measured (Sec. 8).

6 Comparison with micropropulsion

Table 9 compares the device with flown systems in the nanonewton-to-micronewton range. The thrust per watt is about 200 times below an ion thruster, which is Eq. (1) and cannot be engineered away. At micronewtons the power penalty reaches watts and electrospray or FEEP wins. The difference from every flown system is the hardware column: no tank, no feed system, no neutralizer, no total-impulse limit. Nothing in the table produces thrust below 100 nN.

Related concepts share the return-circuit idea. Bare electrodynamic tethers [6, 7] and Janhunen’s electric sail and plasma brake [8, 9] all use continuous electron emission with the ambient plasma closing the circuit, but produce thrust by Lorentz force or ion deflection on kilometre-scale tethers. Atmosphere-breathing electric propulsion [10] collects ambient gas but keeps an ionization stage, an acceleration stage and a neutralizer. A recent review of propellantless propulsion [11] does not list a standalone ionospheric electron-exchange thruster.

7 Bench experiment

The electrode topology was tested in a laboratory vacuum chamber. A W5W automotive bulb filament with the glass removed, held at -50 and -62 V on its two terminals, faces a grounded 4×4 cm copper plate in a glass chamber at 4–5 Pa. The ammeter between plate and earth reads a steady 87 mA once the filament is hot: a single net-negative electrode emits electrons and accelerates them across a gap onto a grounded collector. At this pressure the gap is collisional, so the current includes gas amplification and the result is qualitative. The experiment confirms the principle tested numerically in ladder stage 1: a negative cathode emits and accelerates electrons toward a grounded plate. Controls (bias off, filament cold) and an I–V sweep are the next step.

8 Limitations

Ordered by how much each could change the answer.

1. **Ion clock and reduced ion mass.** Ions are $400 m_e$. The 3D field campaign settled its floats at 6 μ s on that clock and found the 800 ns snapshots at about half the settled float, with thrust moving under 2 %; real O^+ is 73 times heavier still and settles roughly $9 \times$ slower, which no run has reached.
2. **Prescribed beam.** Beam current is a fixed input, not self-consistently emitted from a modelled cathode. The choice of flight emitter is outside the scope of this paper.

Table 9: Position against flight micropropulsion in the nanonewton-to-micronewton range.

system	example	thrust	power	I_{sp} [s]	hardware
this work	electron exchange	3–43 nN	12–278 mW	10^6	HV supply, cathode, aperture
cold gas	GomSpace NanoProp [12]	0.1–10 mN	<2 W	60–110	tanks, valves, nozzles
PPT	PPTCUP [13]	40 μ N	2 W	655	capacitor bank, electrodes
electrospray	Busek CMNT [12]	5–30 μ N	1–16 W	200–300	tanks, PPU, neutralizer
FEEP (1U)	NanoFEEP [2]	0–8 μ N	0.2–0.3 W	6000	liquid-metal needles, heater
FEEP	Enpulsion IFM Nano [14]	10–500 μ N	8–40 W	2000–5000	emitter, 2 neutralizers, PPU

3. **CubeSat sheath regime.** Runs sit at $r/\lambda_D \approx 2.5$; CubeSats are 25–60 λ_D across, where orbital-motion-limited collection no longer applies. A single large-body run would settle this.
4. **Stationary plasma, held attitude.** No ram flow is modelled, and the mission cases assume a fixed pose without an attitude-control model.

All powers are beam power VI ; converter and cathode overheads belong to the flight design.

9 Conclusion

A cathode, a high-voltage supply and the spacecraft’s own skin make a thruster that needs no propellant. PIC simulations measure its operating envelope at 3–43 nN from 12–278 mW, with two constants describing every run to within 3 %, a floating potential fixed by a measured collection law, and a return current that costs 1–9 % of the beam thrust. Against a year of real drag, with the float included, the device holds 500–600 km for 8–39 mW of beam power and covers the 400 km mean demand with a slender body at 140–240 mW. The feasibility condition depends on shape and altitude but not on size, so a 3U CubeSat inherits it at 0.9–3.4 W. The device is cheap to test: it needs no hardware beyond a supply and an electron gun of the kind that already flies as a neutralizer, and it can be retrofitted to an existing ion or Hall thruster after its propellant is spent. The transverse geomagnetic field, once the largest open question, is now measured: at flight strength it leaves the operating point unchanged, and at ten times flight strength it forces field-aligned firing. The flight emitter is the next work.

Data. All simulation inputs, run manifests, gate policies, the orbit propagations and the model are committed in the project repository, archived at DOI:10.5281/zenodo.22115924.

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